

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The total surface area of a cube whose edge is 5 cm is:
(A) (A) 150 cm^2
(B) (B) 125 cm^2
(C) (C) 100 cm^2
(D) (D) 250 cm^2
2. If the slant height of a cone is 10 cm and the radius is 6 cm, its curved surface area is:
(A) (A) $30\pi \text{ cm}^2$
(B) (B) $40\pi \text{ cm}^2$
(C) (C) $50\pi \text{ cm}^2$
(D) (D) $60\pi \text{ cm}^2$
3. The lateral surface area of a cuboid of dimensions $8 \text{ cm} \times 6 \text{ cm} \times 5 \text{ cm}$ is:
(A) (A) 70 cm^2
(B) (B) 110 cm^2
(C) (C) 140 cm^2
(D) (D) 280 cm^2
4. If the probability of an event E occurring is 0.65, what is the probability of the event 'not E '?
(A) (A) 0.25
(B) (B) 0.35
(C) (C) 0.45
(D) (D) 0.65
5. The probability of winning a game is $\frac{7}{12}$. What is the probability of losing the game?

- (A) (A) $1/12$
 (B) (B) $4/12$
 (C) (C) $5/12$
 (D) (D) $7/12$
6. A bag contains balls of different colours. If the probability of picking a red ball is $2/9$, what is the probability that the ball picked is NOT red?
- (A) (A) $1/9$
 (B) (B) $2/9$
 (C) (C) $5/9$
 (D) (D) $7/9$
7. If $P(E) = 0.07$, find $P(\text{not } E)$.
- (A) (A) 0.93
 (B) (B) 0.97
 (C) (C) 0.03
 (D) (D) 0.07
8. A student solves a problem correctly with a probability of 0.85 . What is the probability that the student solves the problem incorrectly?
- (A) (A) 0.05
 (B) (B) 0.10
 (C) (C) 0.15
 (D) (D) 0.25
9. Assertion (A): The total surface area of a solid hemisphere of radius r is $3\pi r^2$. Reason (R): The total surface area of a sphere is $4\pi r^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): If the radius of a cylinder is doubled and its height is halved, the volume remains the same. Reason (R): The volume of a cylinder is given by $V = \pi r^2 h$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

- (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
11. Assertion (A): The volume of a cone with radius r and height h is $\frac{1}{3}\pi r^2 h$. Reason (R): A cone is one-third of a cylinder with the same base and height.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
12. Assertion (A): The slant height l of a cone is given by $l = \sqrt{h^2 + r^2}$. Reason (R): This is derived from Pythagoras theorem applied to the triangle formed by height, radius, and slant height.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
13. Assertion (A): If the surface area of a sphere is 616 cm^2 , its radius is 7 cm. Reason (R): The surface area of a sphere is $4\pi r^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
14. Assertion (A): If the probability of winning a game is 0.7, then the probability of losing the game is 0.3. Reason (R): The sum of the probabilities of all complementary events is 1.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

15. Assertion (A): The probability of an event happening is 0.45, then the probability of the event not happening is 0.65. Reason (R): For any event E, $P(\bar{E}) = 1 - P(E)$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
16. Assertion (A): If a fair coin is tossed, the probability of getting a head is 0.5 and the probability of not getting a head is 0.5. Reason (R): Getting a head and not getting a head are complementary events.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
17. Assertion (A): If the probability of selecting a red ball from a bag is $\frac{2}{7}$, the probability of not selecting a red ball is $\frac{5}{7}$. Reason (R): The probability of an impossible event is 1.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
18. Assertion (A): If $P(E) = 0.12$, then $P(\bar{E}) = 0.88$. Reason (R): $P(E) + P(\bar{E}) = 1$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

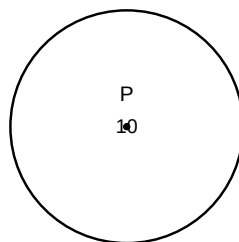
Section B: Very Short Answer (2 marks each)

1. Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

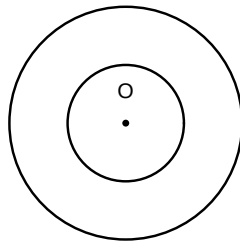
2. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.
3. The slant height of a frustum of a cone is 4 cm and the perimeters of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.
4. A solid is in the form of a right circular cone mounted on a hemisphere. If the radius of the hemisphere and the cone is 3.5 cm and the height of the cone is 4 cm, find the volume of the solid. (Use $\pi = 22/7$)
5. The diameter of a metallic ball is 4.2 cm. What is the mass of the ball, if the density of the metal is 8.9 g/cm^3 ?
6. In $\triangle ABC$, a circle is inscribed touching sides AB , BC , and CA at points P , Q , and R respectively. If $AB = 12 \text{ cm}$, $BC = 8 \text{ cm}$, and $AC = 10 \text{ cm}$, find the length of AP .

R
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7. From a point P , 10 cm away from the centre of a circle, the length of the tangent drawn to the circle is 8 cm. Find the radius of the circle.



8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.



9. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , find $\angle POA$.

P
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Section C: Short Answer (3 marks each)

1. A metallic sphere of radius 6 cm is melted and recast into the shape of a cylinder of radius 4 cm. Find the height of the cylinder.
2. Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
3. A conical tent is 10 m high and the radius of its base is 24 m. Find the slant height of the tent and the cost of the canvas required to make the tent, if the cost of 1 m^2 canvas is Rs. 70.
4. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.
5. A hemispherical tank, full of water, is emptied by a pipe at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to empty half the tank, if it is 3 m in diameter? (Take $\pi = \frac{22}{7}$)

Section D: Long Answer (4-5 marks each)

1. A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. If a right circular cylinder circumscribes the toy, find the difference between the volume of the cylinder and the volume of the toy. (Take $\pi = 3.14$)
2. A metallic sphere of radius 6 cm is melted and recast into the shape of a right circular cylinder of radius 6 cm. Find the height of the cylinder.
3. A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm^3 . The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket. (Use $\pi = 3.14$)
4. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.
5. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number, (ii) a perfect square number, (iii) a number divisible by 5. Use the concept of complementary events to verify the probability of the event 'not getting a perfect square'.
6. A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game using the complementary event property.
7. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls in the bag. Further, if a ball is drawn at random, what is the probability of not drawing a red ball?
8. Two dice are thrown simultaneously. Find the probability of getting (i) a sum of 8, (ii) a doublet, (iii) a sum greater than 9. Use the complementary event concept to find the probability of getting a sum not equal to 8.